Prometheus Alerts in Maquette

Prometheus

Prometheus Monitoring is used to monitor the health of fraud api. This video is good resource to get how it works : video.

This is an example of how to create a alert :

```
{
  prometheusAlerts+:: {
    groups+: [
      {
        name: 'ocl-maquette.rules',
        rules: [
          {
            alert: 'MaquetteP99LatencyHighForOEDomain',
            expr: 'histogram_quantile(0.99, sum(rate(fraudcheck_request_response_time_seconds_bucket{ domain =
"OE" }[20m])) by (prometheus_cluster, domain, le)) * 1000 > 2000',
            'for': '15m',
            labels: {
              severity: 'critical',
            },
            annotations: {
              summary: 'Maquette P99 latency is over 2s for {{ $labels.domain }} on {{ $labels.
prometheus_cluster }} in the past 15 minutes',
              description: 'Maquette latency is over 2s in the past 15 minutes, please check Cassandra/ Model
or Maquette process for issue.',
            },
          },
        ],
      },
    ],
  },
}
```

The code snippet demonstrates a configuration for Prometheus alerts. Below is the explanation of various parameters from this snippet.

- `prometheusAlerts+::`: This is the top-level field and represents a set of Prometheus alerts. The `+::` operator denotes that the field is being merged with an existing configuration, if any.
- `groups+: []`: This field defines a list of alert groups. Alert groups are used to organize related alerts together.
- `name: 'ocl-maquette.rules'`: This field specifies the name of the alert group. In this case, the group is named 'ocl-maquette.rules'.
- `rules: []`: This field represents a list of alert rules within the group.
- `alert: 'MaquetteP99LatencyHighForOEDomain'`: This field defines the name of the alert. The alert is named 'MaquetteP99LatencyHighForOEDomain'.
- `expr: 'histogram_quantile(0.99, sum(rate(fraudcheck_request_response_time_seconds_bucket{ domain = "OE" }[20m])) by (prometheus_cluster, domain, le)) * 1000 > 2000'`: This field specifies the Prometheus expression that determines when the alert should fire. The expression calculates the 99th percentile latency for requests with the domain "OE" over a 20-minute window, and checks if it exceeds 2000 milliseconds.
- `'for': '15m'`: This field specifies the duration for which the expression should be evaluated to determine if the alert should fire. In this case, it is set to 15 minutes.
- `labels: { severity: 'critical' }`: This field defines the labels associated with the alert. Labels provide additional metadata to the alert. In this case, the severity label is set to 'critical'.
- `annotations: { summary: '...', description: '...' }`:The 'summary' annotation provides a summary of the alert, including placeholders for labels, and the 'description' annotation provides a detailed description of the alert.

fraudcheck_request_response_time_seconds_bucket

It is a prometheus histogram metric. It measures the fraud check requests in seconds. A histogram groups observations into configurable buckets and counts the number of observations that fall into each bucket. Each bucket is represented by a time series with the suffix `_bucket` and a label `le` that specifies the upper bound of the bucket.

You can learn about Prometheus Histogram from here.

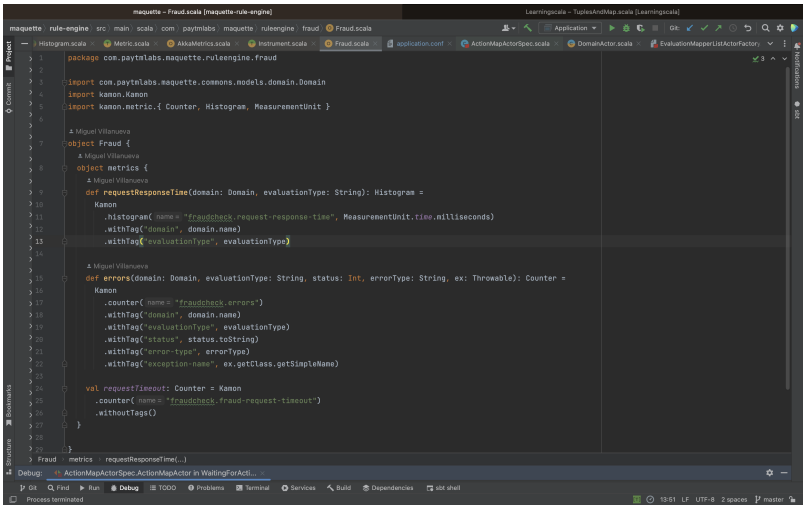
Example of time series from the histogram is :

```
fraudcheck_request_response_time_seconds_bucket{container="rule-engine", domain="BOSS", endpoint="metrics", evaluationType="maker_checker", instance="rule-engine-rollout-7b6cdc9cff-4mp6x", job="maquette/rule-engine", le="0.01", namespace="maquette", pod="rule-engine-rollout-7b6cdc9cff-4mp6x", rollout_pod_type="stable", service="rule-engine-service"}
```

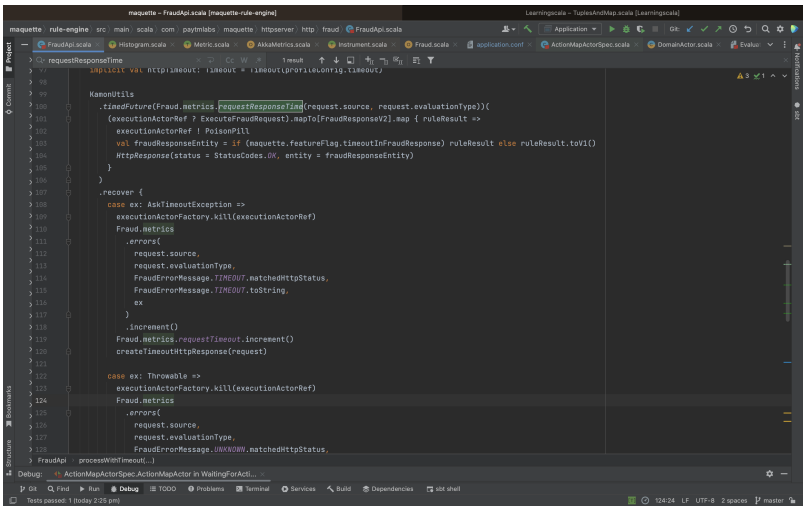
- `container`: The name of the container where the metric was collected, in this case `rule-engine`.
- `domain`: The "domain" label likely represents the domain or area of the system where the metric is being measured, in this case `BOSS`.
- `endpoint`: The "endpoint" label indicate the specific endpoint or API being monitored, in this case `metrics`
- `evaluationType`: The type of evaluation used for the fraud check requests, in this case `maker_checker`.
- `instance`:
- `job`: The job associated with the metric, in this case `maquette/rule-engine`.
- `le`: The upper bound of the bucket, in this case `0.01`.
- `namespace`: The "namespace" label typically refers to a namespace in which system or application is running. , in this case `maquette`.
- `pod`:
- `rollout_pod_type`:
- `service`: The service associated with the metric, in this case `rule-engine-service`.

Below are the code snippets to configure metric :

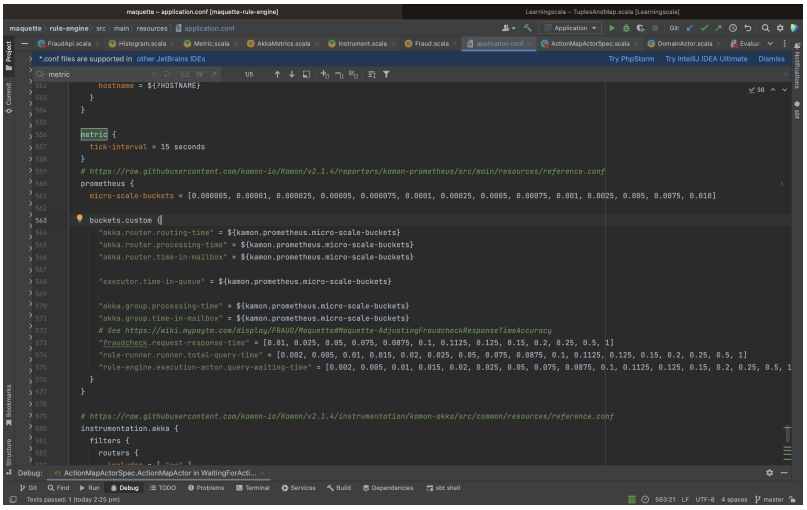
1. Below snippet is used to initialise the metric :



2. below snippet shows how to use the metric once api is called:



3. Configuration can be set up with customised bucketing of time :



You can learn more about alerts in prometheus from here.

This are the PRs for Risk execution Timeout Management :

1. https://bitbucket.org/paytmteam/pai-kubernetes-deployments/pull-requests/9342
2. https://bitbucket.org/paytmteam/pai-kubernetes-deployments/pull-requests/9372